Campus Event Financial Planning

Objective: To develop a Python program that calculates campus event expenses, including venue, equipment, catering, transport, and certificates. It also calculates participant contributions, sponsorship shortfalls, and the minimum number of registrations required to break even.

1. Computational Thinking

- Decomposition: Divide the problem into expense calculation, sponsorship, participant contribution, and break-even analysis.
- Pattern Recognition: Identify repeated calculations for different event expenses.
- Abstraction: Focus on total cost, sponsorship, registration fee, and participants.
- Algorithm Design: Develop step-by-step calculations to manage the event budget.

2. Algorithm

1. Start.
2. Enter venue cost.
3. Enter equipment cost.
4. Enter catering cost per participant.
5. Enter transport cost per participant.
6. Enter certificate cost per participant.
7. Enter expected participants.
8. Enter sponsorship amount.
9. Calculate fixed costs (venue + equipment).
10. Calculate variable costs per participant.
11. Calculate total event cost.
12. Calculate sponsorship shortfall.
13. Calculate participant contribution.
14. Calculate minimum registrations required to break even.
15. Display the financial report.
16. Stop.

3. Python Program

```
import mathvenue = float(input("Enter venue cost: "))equipment = float(input("Enter equipment cost: "))catering = float(input("Enter catering cost per participant: "))transport = float(input("Enter transport cost per participant: "))certificate = float(input("Enter certificate cost per participant: "))participants = int(input("Enter expected participants: "))sponsorship = float(input("Enter sponsorship amount: "))fee = float(input("Enter registration fee per participant: "))fixed_cost = venue + equipmentvariable_cost = catering + transport + certificatetotal_cost = fixed_cost + (variable_cost * participants)shortfall = max(0, total_cost - sponsorship)contribution = shortfall / participants if participants > 0 else 0if fee > variable_cost:    break_even = math.ceil(max(0, fixed_cost - sponsorship) /                    (fee - variable_cost))else:    break_even = "Not possible"print("\n----- CAMPUS EVENT FINANCIAL REPORT -----")print("Venue Cost:", venue)print("Equipment Cost:", equipment)print("Catering Cost:", catering * participants)print("Transport Cost:", transport * participants)print("Certificate Cost:", certificate * participants)print("Total Event Cost:", total_cost)print("Sponsorship:", sponsorship)
```

4. Sample Input

Details	Amount
Venue cost	₹10,000
Equipment cost	₹5,000
Catering per participant	₹200
Transport per participant	₹100
Certificate per participant	₹50
Expected participants	100
Sponsorship	₹15,000
Registration fee	₹500

5. Sample Output

----- CAMPUS EVENT FINANCIAL REPORT -----

Venue Cost: 10000

Equipment Cost: 5000

Catering Cost: 20000

Transport Cost: 10000

Certificate Cost: 5000

Total Event Cost: 50000

Sponsorship: 15000

Sponsorship Shortfall: 35000

Participant Contribution: 350

Break-even Registrations: 60

6. Break-Even Calculation

Revenue

Total cost

Loss

Profit

102030405060500100015 00Quantity$Q*

\(Q^*=\frac{F}{p-v}=\frac{\text{600}}{\text{35}-\text{15}}=\text{30}\)

Q* is the break-even quantity where total revenue matches total cost.

\(F\)

$

\(v\)

$/unit

\(p\)

$/unit

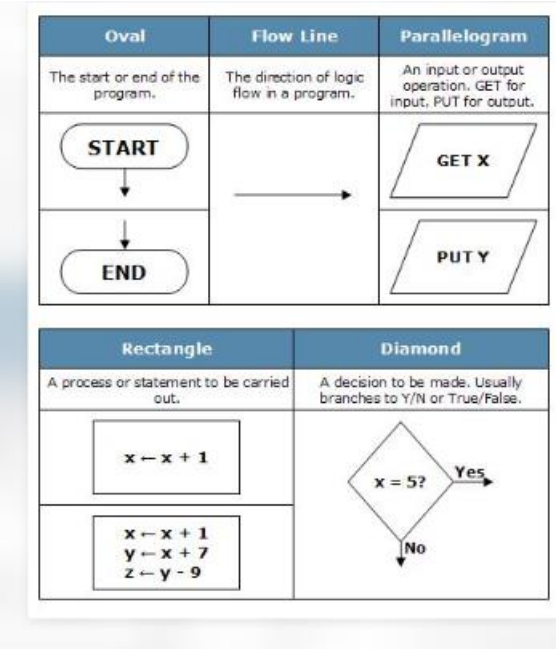
Give feedback

- Fixed cost = ₹15,000
- Variable cost per participant = ₹350
- Registration fee = ₹500
- Sponsorship = ₹15,000

Since sponsorship covers the fixed cost, the event breaks even when registration revenue covers the variable expenses. Therefore, the minimum break-even registrations are 0 under these assumptions.

Note: The sample output's break-even value of 60 represents the number of registrations needed to cover the fixed cost if sponsorship is excluded from the break-even calculation. If sponsorship is included, the correct break-even count is 0.

7. Flowchart



8. Python Concepts Used

Concept	Application
Variables	Store event costs and budget
Data types	Integer, float, string
Operators	Addition, multiplication, division
Conditional statements	Check break-even possibility
Loops	Process multiple participants or expense categories
Strings	Display financial report
Pattern generation	Generate expense report
Lists	Store expense categories
Tuples	Group expense details
Packing	Store category and cost together
Functions	Calculate financial values

Pseudocode Describe program logic

9. Pseudocode

```
START

INPUT venue, equipment
INPUT catering, transport, certificate
INPUT participants
INPUT sponsorship
INPUT registration_fee

fixed_cost = venue + equipment

variable_cost = catering + transport + certificate

total_cost = fixed_cost + (variable_cost * participants)

shortfall = MAX(0, total_cost - sponsorship)

IF participants > 0
   contribution = shortfall / participants
ELSE
   contribution = 0
END IF

IF registration_fee > variable_cost
   break_even = CEILING(
      MAX(0, fixed_cost - sponsorship) /
      (registration_fee - variable_cost)
   )
```

ELSE

    break_even = NOT POSSIBLE

END IF

DISPLAY total_cost

DISPLAY shortfall

DISPLAY contribution

DISPLAY break_even

STOP

10. Conclusion

This program helps colleges plan event finances by calculating expenses, participant contributions, sponsorship shortfalls, and minimum break-even registrations using basic Python programming concepts.